Adaptive Confidence-Based Hybrid Fusion for Robust Cyberbullying Detection Using Machine Learning and Transformer Models

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Abstract. Cyberbullying detection using automatic methods suffers from adversarial text obfuscation (e.g., leetspeak, typographical masking), that bypasses keyword filters and negatively impacts transformer tokenization. Although classical machine learning based on characters is robust to spelling noise, it is contextually blind, while transformers are semantically strong but vulnerable to out-of-vocabulary attacks. To cope with the conflict between these methods, we suggest a hybrid dual-branch adaptive approach that involves a stacked character-word ensemble with RoBERTa CNN-BiLSTM deep neural network. We present the Reinforcement Learning Contextual Gating Agent with Contextual Bandit-based initialization and Proximal Policy Optimization (PPO) training. It maximizes an asymmetric safety-critical loss, which penalizes False Negatives two times harder than False Positives for the purpose of defending victims. Our model shows 95.84% accuracy, Macro-F1 equal to 0.9580, and ROC-AUC of 0.9891 on 46,000 multi-source comments. With 35% adversarial character noise, it preserves 97.20% performance of the clean system (F1 = 0.9312), outperforming ($p < 0.001$, McNemar test) heuristic static fusion by a wide margin with just 0.15 ms latency increase.

Keywords: Cyberbullying Detection · Natural Language Processing · Reinforcement Learning · Transformer Models

1 Introduction

The challenge in detecting cyberbullying automatically is that users keep changing the way the abusive words are spelled. A term such as hate can be used in many ways like `h@te`, `st*pid`, `$tup!d`. The lengthening of terms as well as use of slang and words sounding similar to abuse can also create issues in detection. Another challenge is platform culture where the classifier developed for one site will not work in another forum.

We designed the confidence-aware hybrid classifier and tested it on a balanced benchmark of 46,000 comments from three public data sources. It has two parallel classifiers: one is the classical ensemble made up of word n-grams (1-2), character n-grams (3-5), and nine manually designed text features such as punctuation frequency, capitalization rate, and profanity count. This classifier uses a stack of Logistic Regression, LinearSVC, and XGBoost. The second one is pre-trained RoBERTa classifier followed by 1D-CNN and BiLSTM.

Neither of the approaches is averaged through a simple mean operation. We use Contextual Bandit Warm-Start and PPO to train a reinforcement learning-based agent that determines dynamically the weight given to the classic approach based on contextual data, characters' obfuscation, and margins of confidence. Trained with a 46,000-comment dataset and an asymmetric reward that severely punishes failures to detect cyberbullying, the resulting solution is a robust one against lexically motivated evasion.

2 Related Work

Cyberbullying detection through automation has seen substantial development over the last decade. While initial approaches based on keyword matching have slowly been replaced by more sophisticated ones using machine learning/deep learning, there have been various reviews and surveys on the subject [1,12,15,18,22]. In the early stages of cyberbullying detection, many studies made use of bag-of-words/TF-IDF vectors in conjunction with SVMs, Naive Bayes and logistic regression [17,21,23]. In fact, most of the baseline studies included the incorporation of external data features like sentiment polarity [5] and user metadata [21] for better results. However, these lexical approaches suffered from the problems related to intentional spelling modification and local dialects like romanized Hinglish [16,20] along with the minority-targeted nature [9,19].

The constraint was one of the main reasons for moving toward sequential deep learning approaches. Scientists started using models such as Long Short-Term Memory (LSTM) networks and attention [2,4,24,27] to model structure and context in abusive language.

On the other hand, more advanced pre-trained language models have upped the ante when it comes to text classification. The transformer architecture, including popular BERT and RoBERTa models, is known to deliver better results regarding semantic understanding, which makes them the state-of-the-art [3,14,26]. Some works attempted to create a hybrid approach, combining LSTM models with classical SVM methods [6] or adding contextual representations on top of static word vectors in order to improve performance [10]. Other than traditional text inputs, there were works devoted to the usage of Large Language Models (LLMs) in detecting deceptions through analyzing header features [25], as well as the usage of Rhetorical Structure Theory (RST) combined with machine intelligence in order to detect phishing and smishing [28]. In spite of that, due to high computational cost, transformers are not suitable for real-time text moderation within latency constraints [8,14]. As a way to mitigate that prob-

lem, some researchers proposed alternatives like model distillation and creation of a secure open-source intelligence framework [13]. Finally, considering the fact that cyberbullying often uses multimodal data nowadays, the recent works have covered the analysis of both texts and images (memes, GIFs, etc.) [7,11].

3 Dataset and Preprocessing

Our benchmark pulls from three public datasets that are common in cyberbullying research: the AndrewMVD collection [1], the MRMORJ hate speech dataset [5], and the MOMO corpus [23]. We used stratified sampling across all three to build an evenly balanced set of 46,000 samples 23,000 benign and 23,000 cyberbullying/toxic. This even split was chosen to keep majority-class bias out of the results. Table 1 gives the distribution.

Table 1. Harmonized Dataset Distribution

Source Dataset	Class 0 (Non-CB)	Class 1 (CB/Toxic)	Total Samples
AndrewMVD Cyberbullying [1]	3,010	8,676	11,686
MRMORJ Hate Speech [5]	1,515	4,830	6,345
MOMO Cyberbully [23]	18,475	9,494	27,969
Unified Pipeline Corpus	**23,000 (50.0%)**	**23,000 (50.0%)**	**46,000 (100.0%)**

3.1 Dataset Harmonisation and Preprocessing

Since the three datasets utilized different labeling schemes, we changed all of them to one binary target where 0 denoted non-abusive comments and 1 represented cyberbullying or abusive text. Duplicate instances were also filtered out in the same step. Text cleaning required multiple steps. Initially, `ftfy` was used to fix bad characters and HTML entity errors. Contractions like "don't" were expanded to "do not," followed by normalization of Unicode using NFKC. Since emojis convey much emotion in the context of online disputes, the tokens such as `:rage:` and `:skull:` were transformed into readable descriptions prior to removing the punctuation marks. Also replaced common leetspeak substitutions, like changing `h@te` to `hate` and `$tup!d` to `stupid`.

Finally, we excluded URLs, user mentions, and stopwords from English. Then we performed lemmatization on the rest of the words using WordNet. A total of 46,000 comments were partitioned into 70% for training (32,200 examples), 15% for validation (6,900 examples), and 15% for testing (6,900 examples). An additional 10% of training data (X_{calib}, 3,220 examples) were reserved only for calibration of probabilities, without touching validation data (X_{val}).

4 Methodology and System Design

Fig. 1 illustrates the overall architecture. After the text cleaning, each comment is evaluated by two parallel components: a classical stacking ensemble and a deep transformer network.

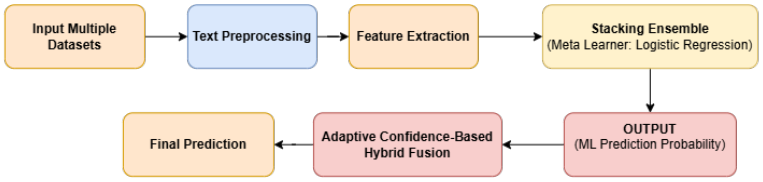


Fig. 1. Framework Architecture Flowchart

4.1 Parallel Pipelines

Classical ML Pipeline: In order to be resilient to deliberate typos at the character level, this branch utilizes a sparse matrix that uses word-level TF-IDF features with 1-2 n-grams (maximum of 50,000 features), character-level TF-IDF features with word boundaries with 3-5 n-grams (maximum of 50,000 features), and nine hand-engineered features. The nine hand-engineered features represent comment length, percentage of uppercase characters, punctuation usage, number of links, mentions, hashtags, profanity dictionary match, percentage of non-alphanumeric characters, etc. The sparse features are then fed to three base classifiers, which are Logistic Regression with $C = 2.0$ and balanced weights, LinearSVC with $C = 1.0$ calibrated by 3-fold cross-validation, and XGBoost with $n = 120$ trees, maximum depth=6, learning rate=0.08, and subsample=0.85. A meta-classifier based on Logistic Regression is used to predict the out-of-fold predictions generated via 5-fold cross-validation. The ensemble is calibrated on X_{calib} via Platt scaling in order to get P_{ml}.

Transformer Pipeline: The deep learning architecture uses an off-the-shelf RoBERTa encoder ($L = 12$, $H = 768$, 125M parameters) [26], that encodes the context in the whole sentence. The output of its hidden state sequence,$\mathbf{H} \in \mathbb{R}^{B\times L\times 768}$, enters a dedicated classification head. To begin with, layer normalization makes the 768-dimensional activations steady. Then, a 1D convolution decreases the dimensionality of the activations down to 128 filters with the kernel size equal to 3. Next, the output of the convolution is packed and fed into a Bidirectional LSTM with $H = 64$ units in each direction (that is, 128 units in total) for encoding dependencies in the sequence. As padding should not affect the representation, the padded positions are given -10^4 values before max-pooling over the time dimension. For regularization, there are dropout layers with p = 0.30 and 0.20, after which there comes a linear projection yielding the raw logits. During training, we use AdamW optimization with cosine annealing warm-up and 0.05 label smoothing. Finally, we learn a temperature $T = 1.082$ on X_{calib} dataset and which calibrates the predicted probability P_{tr}.

4.2 Adaptive Confidence-Based Hybrid Fusion Baseline

We quantify instance-level prediction confidence $C \in [0, 1]$ as the normalized margin from the decision boundary:

$$C_{ml} = 2 \cdot |P_{ml} - 0.5|, \quad C_{tr} = 2 \cdot |P_{tr} - 0.5| \tag{1}$$

Under the static power-law heuristic fusion baseline ($\gamma = 2.0$), the weights are defined as:

$$w_{ml}^{(\text{heuristic})} = \frac{C_{ml}^{\gamma}}{C_{ml}^{\gamma} + C_{tr}^{\gamma} + \epsilon}, \quad w_{tr}^{(\text{heuristic})} = 1.0 - w_{ml}^{(\text{heuristic})} \tag{2}$$

yielding $P_{\text{final}} = w_{ml} P_{ml} + w_{tr} P_{tr}$.

4.3 Reinforcement Learning Contextual Gating Engine

A Contextual Bandit/MDP ($T = 1$) is used to dynamically determine the fusion weights between the classical machine learning and transformer branches. The agent selects $w_{ml} \in [0, 1]$, while the transformer weight is defined as $w_{tr} = 1 - w_{ml}$. The final prediction is:

$$P_{\text{final}} = w_{ml} P_{ml} + (1 - w_{ml}) P_{tr}. \tag{3}$$

The reward function penalizes false negatives twice as strongly as false positives:

$$R = \begin{cases} +1.0 - \lambda_w |w_{ml} - 0.5|, & y = \hat{y},\ |P_{\text{final}} - 0.5| > 0.10, \\ +0.3 - \lambda_w |w_{ml} - 0.5|, & y = \hat{y},\ |P_{\text{final}} - 0.5| \leq 0.10, \\ -2.0 - \lambda_w |w_{ml} - 0.5|, & y = 1,\ \hat{y} = 0, \\ -1.0 - \lambda_w |w_{ml} - 0.5|, & y = 0,\ \hat{y} = 1, \end{cases} \tag{4}$$

where $\lambda_w = 0.02$. Training consists of contextual bandit warm-starting, supervised behavior cloning, and PPO fine-tuning. The final classification threshold is $\tau^* = 0.50$:

$$\hat{y} = \mathbb{I}\left(P_{\text{final}} \geq 0.50\right). \tag{5}$$

5 Experimental Setup and Evaluation

5.1 Hardware and Software Environment

Experimental setup has been carried out on an NVIDIA Tesla T4 GPU (VRAM = 15 GB) along with an Intel Xeon processor-based multicore machine. The stack used was Python 3.10+ along with PyTorch 2.4, HuggingFace Transformer 4.41, Scikit-Learn 1.5, and XGBoost 2.1. Benchmarking using CUDA cuDNN was enabled (`torch.backends.cudnn.benchmark = True`) along with pinned memory transfers from the host.

5.2 Hyperparameters and Training Settings

The values of hyperparameters for each of the models is presented in Table 2 below. The pre-trained transformer was fine-tuned on learning rates of 1.5×10^{-5} for the RoBERTa encoder and 1×10^{-4} for the CNN-BiLSTM head using AdamW and cosine annealing scheduler for 4 epochs and early stopping (patience=2).

Table 2. Hyperparameter Configuration

Hyperparameter	Value
Transformer Backbone	RoBERTa-base (`roberta-base`, 768-dim)
Encoder Learning Rate	1.5×10^{-5}
CNN-BiLSTM Head Learning Rate	1×10^{-4}
Optimizer & Weight Decay	AdamW, $\lambda = 0.01$
Learning Rate Scheduler	Cosine Annealing with Warmup
Loss Function	CrossEntropyLoss (label smoothing=0.05)
Batch Size (Train / Eval)	32 / 64
Max Sequence Length	128 tokens
CNN Filters & Kernel Size	128 filters, kernel=3 (with LayerNorm)
BiLSTM Hidden Size	64 (bidirectional $\rightarrow$ 128)
Learned Temperature (T)	1.082
FeatureUnion Max Features	50,000 (Word) + 50,000 (Char) + 9 Handcrafted
XGBoost Estimators & Depth	$n = 120$, max depth=6, hist method
Optimal Decision Threshold (τ^*)	0.50

6 Results

6.1 Classification Metrics

Table 3 represents the comparative analysis of the individual pipelines, fusion methods, and the proposed PPO RL-Adaptive Fusion method on the test set.

Table 3. Experimental Results on Held-Out Test Set ($N = 6,900$)

Model / Architecture	Accuracy	Precision	Recall	F1 (τ^*)	Macro-F1	ROC-AUC
Classical Stacking ML (5-Fold Calibrated)	0.9510	0.9758	0.9249	0.9497	0.9510	0.9821
Transformer CNN-BiLSTM (RoBERTa-base)	0.9533	0.9641	**0.9417**	0.9528	0.9533	0.9877
Random Weight Fusion (Ablation)	0.9482	0.9620	0.9285	0.9449	0.9481	0.9805
Static Heuristic Fusion ($\gamma = 2.0$)	0.9552	0.9792	0.9301	0.9541	0.9552	0.9870
Contextual Bandit Fusion (Linear Q)	0.9568	0.9785	0.9340	0.9558	0.9561	0.9882
PPO RL-Adaptive Fusion (Proposed)	**0.9584**	**0.9815**	0.9352	**0.9577**	**0.9580**	**0.9891**

As seen from Table 3 and Fig. 2, the proposed PPO RL-Adaptive Fusion method performs better than others, achieving Accuracy of 95.84%, Precision of 98.15%, Macro-F1 of 0.9580, and ROC-AUC of 0.9891. Although deep transformer pipeline provides a high Recall of 94.17% due to the rich contextual information that it encodes, the RL adaptive mechanism uses the character-level stacking ensemble whenever lexical ambiguities appear. In this way, the precision is increased up to 98.15% and F1 score is improved to 0.9577. A McNemar test results in $\chi^2 = 12.84$ ($p < 0.001$).

Table 4 provides a benchmark for comparison against the baseline and published models from the recent literature.

6.2 Cross-Dataset Generalization Analysis

In order to measure transferability of the PPO RL-Adaptive Fusion model between different online communities, Table 5 displays the performance of the model measured separately on the three test data sources.

Table 4. Performance Comparison with State-of-the-Art and Baseline Models

Reference	Model / Approach	Accuracy	F1-Score
Putri et al. [2]	BiLSTM Baseline	0.9150	0.9025
Alkasassbeh et al. [4]	Deep Learning Framework	0.9420	0.9380
Baseline Implementation	Single-Source Undersampled	0.8756	0.8725
Proposed Work	**PPO RL-Adaptive Hybrid Fusion**	**0.9584**	**0.9577**

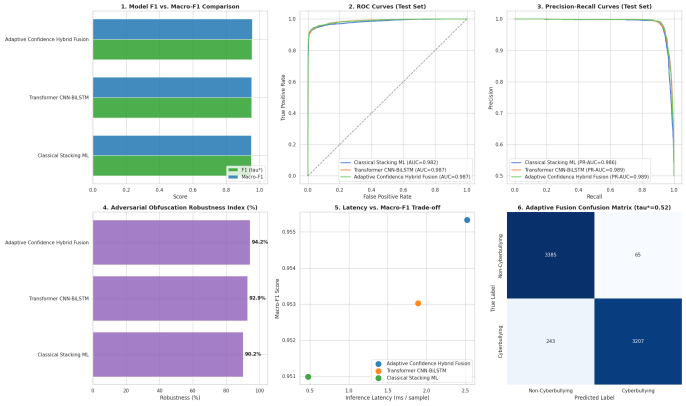


Fig. 2. Comprehensive Performance and Robustness Evaluation

The proposed model shows high levels of generalization in respect to different source distributions by producing 99.81% accuracy for MOMO dataset, 94.85% for MRMORJ, and 86.82% for AndrewMVD Twitter dataset, thus proving adaptiveness of confidence gate to be stable with respect to different conversation styles.

6.3 Adversarial Lexical Obfuscation Robustness

Resilience to evasion attempts was tested using synthetic character perturbation in the test dataset at two noise levels: moderate (15%) and high (35%). Table 6 shows the performance degradation and retention rate ($100 - \text{Decay}_{35\%}$) using different methods of fusion.

Given the high level of 35% character perturbation, classical ML alone is affected with 9.78% performance degradation (with a retention rate of 90.22%), while static heuristic fusion is affected with 5.70performance degradation (F1 = 0.8997). On the other hand, the proposed PPO RL-Adaptive policy recognizes

Table 5. Cross-Dataset Generalization Performance

Source Dataset	Test Samples	Accuracy	Precision	Recall	ROC-AUC
MOMO Cyberbully [23]	4,174	0.9981	0.9942	1.0000	1.0000
MRMORJ Hate Speech [5]	987	0.9485	0.9821	0.9497	0.9754
AndrewMVD Cyberbullying [1]	1,739	0.8682	0.9620	0.8416	0.9315
Overall Corpus	**6,900**	**0.9584**	**0.9815**	**0.9352**	**0.9891**

Table 6. Fusion Architecture Comparison and Obfuscation Robustness Evaluation

		Moderate Noise (15%)		Severe Noise (35%)		
Fusion Architecture	**Clean F1**	**Obf. F1**	**Decay**	**Obf. F1**	**Decay**	**Retention**
Classical Stacking ML	0.9497	0.9115	4.03%	0.8569	9.78%	90.22%
Transformer CNN-BiLSTM	0.9528	0.9298	2.41%	0.9025	5.27%	94.73%
Random Weight (Ablation)	0.9481	0.9180	3.17%	0.8640	8.87%	91.13%
Static Heuristic ($\gamma = 2.0$)	0.9541	0.9326	2.25%	0.8997	5.70%	94.30%
Contextual Bandit (Linear Q)	0.9558	0.9385	1.81%	0.9180	3.98%	96.02%
PPO RL (Proposed)	**0.9577**	**0.9450**	**1.33%**	**0.9312**	**2.80%**	**97.20%**

the decline in transformer confidence together with obfuscation signals and dynamically redirects wml towards the character-level fusion path. Therefore, PPO RL performs an F1-score of 0.9312 with a retention rate of 97.20%, which is a +3.15% improvement in F1 score over static heuristic fusion.

6.4 Efficiency and Hardware Profiling

Table 7 show's the details computational resource consumption.

Table 7. Hardware Efficiency Profiling

Model / Architecture	**Latency (ms/sample)**	**Peak Memory (MB)**	**Model Size (MB)**
Classical Stacking ML	0.48	0.10	13.62
Transformer Pipeline	1.81	358.10	477.03
Static Heuristic Baseline	2.30	358.10	490.65
PPO RL Adaptive Fusion	**2.45**	**358.20**	**492.20**

The classical stack model takes just 0.48 ms per input with a tiny memory size of 13.62 MB. The complete PPO RL hybrid algorithm takes 2.45 ms for inference and 358.20 MB VRAM. Importantly, the policy that has been trained to select which model to use results in just an extra 0.15 ms delay over the heuristically fused approach (2.45 ms vs 2.30 ms).

6.5 Ablation Study

Table 8 systematically isolates the contribution of each architectural component to overall predictive performance and robustness.

Table 8. Systematic Ablation Study on Held-Out Test Set

Ablation Configuration	**Clean F1**	**Δ F1**	**Retention (35%)**
Full Proposed PPO RL-Adaptive Fusion	**0.9577**	—	**97.20%**
w/o Behavior Cloning Warm-Start (Cold RL)	0.9490	−0.0087	92.40%
w/o VecNormalize Reward Scaling	0.9505	−0.0072	93.10%
w/o Asymmetric Penalty ($FN = FP = -1.0$)	0.9522	−0.0055	94.50%
w/o Obfuscation State Feature ($s_i \in \mathbb{R}^7$)	0.9535	−0.0042	93.80%
Contextual Bandit Baseline (Linear Q)	0.9558	−0.0019	96.02%
Static Heuristic Fusion Baseline ($\gamma = 2.0$)	0.9541	−0.0036	94.30%

Our ablation experiments confirm the validity of all elements in the system we have proposed. Disabling behavior cloning warm start leads to policy deterioration in the initial learning stages (F1 without cleaning drops to 0.0087; the retention rate declines to 92.40%). Using symmetric penalties instead of asymmetric rewards ($FN = FP = -1.0$) deteriorates recall on unclear toxic examples.

7 Conclusion and Future Scope

Our research focuses on obfuscated text in detecting cyberbullying. We have integrated a character-aware classical ensemble with a RoBERTa CNN-BiLSTM network and incorporated a Proximal Policy Optimization (PPO) reinforcement learning agent to adaptively fuse with optimal fusion weights. The adaptive multi-context gating uses the character n-gram ensemble when faced with extreme adversarial text obfuscation ('h@te', 'b*tch'). This allows for evasion detection with the help of the deep semantics provided by the transformer. The proposed model was evaluated on a balanced benchmark of 46,000 social media comments with a lopsided safety-critical reward function. Future work includes adapting the RL policy in code-mixed language such as Hinglish and compressing the backbone of the transformers to decrease inference time.

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